✅ What Are Proxies in SQL Server Agent?

A **SQL Server Agent Proxy** allows SQL Server Agent jobs to run **specific job steps** under a **different security context** than the SQL Server Agent service account.

In simple terms:
A **Proxy** = a controlled way of allowing job steps to run with a specific set of Windows/SQL permissions without giving full server access.

A proxy is associated with a **Credential**, which stores the security identity (Windows user, Managed Service Account, etc.).
Then, specific job steps (PowerShell, SSIS, CmdExec, T-SQL, etc.) can run using that identity.

📦 Key Components of a Proxy

Component	Description
Credential	Stores a Windows/AD account username & password. Proxies map to these.
Proxy	Grants access to one or more SQL Agent subsystems (CmdExec, SSIS, PowerShell, etc.)
Subsystems	Types of job steps (PowerShell, SSIS package execution, CmdExec, etc.)
SQL Logins / Roles	Access is granted to selected SQL logins that may use the proxy for job steps.

🛠️ Why Do We Need Proxies?

SQL Server Agent by default runs under its service account.
However:

- You **should not** give the service account unnecessary OS-level permissions.
- Sometimes, job steps require **file system access**, **PowerShell script execution**, or **network share access**, which the service account should not have.
- Different teams require different permissions to run their scheduled tasks.

Thus, **Proxies provide least-privilege and controlled execution**.

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⚙️ What Can a Proxy Run? (Subsystems)

A proxy can be assigned to specific job step types like:

- **CmdExec** (batch commands)
- **PowerShell**
- **SSIS** (Integration Services)
- **Operating System (CmdExec)**
- **Replication tasks**
- **SQL Server Analysis Services**
- **Azure storage tasks**

🚀 Real-Time Use Cases & Scenarios

Below are practical scenarios from real SQL Server environments.

🔶 Scenario 1: Backup Files to a Network Share

Problem

A SQL Agent job must copy backup files from local disk to:
\\BackupServer\NightlyBackups
The SQL Server Agent service account **does not** have permission to this network share.

Solution Using Proxy

1. Create a **Windows credential** using a domain account that *does* have access to the share.
2. Create a Proxy for **Operating System (CmdExec)**.

3. Create a job step using robocopy under that proxy.

Benefits

- SQL Agent account remains locked down.
- Only that job step gets share access.

◆ Scenario 2: Running PowerShell Scripts for Maintenance

Problem

A DBA needs a scheduled job to:

- Archive old log files
- Create folders
- Use PowerShell modules (filesystem, network commands)

But SQL Server Agent account lacks filesystem privileges.

Solution Using Proxy

Create a Proxy for **PowerShell** using a credential mapped to a Windows admin account.

Example Job Step

```
Get-ChildItem "D:\Logs" -Recurse | Where-Object {$_.LastWriteTime -lt (Get-Date).AddDays(-15)} | Remove-Item
```

Benefit

PowerShell runs with needed rights without raising SQL Server service privileges.

◆ Scenario 3: Running SSIS Packages That Access File System/FTP

Problem

An SSIS package downloads files using FTP and writes them to folders requiring high-level OS permissions.

SQL Agent job running SSIS cannot execute because of permission issues.

Solution

Create a Proxy mapped to a domain service account with required access.

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Benefits

- The Proxy handles the SSIS task permissions.
- SQL Server Agent remains minimal-privilege.

◆ Scenario 4: Developer Needs to Run a Job Step with External Executables

A developer needs to run a python.exe or .exe for data ingestion.

But you don't want to give the developer:

- sysadmin
- SQL Agent job owner rights
- OS permissions

Solution

1. Create a **Proxy** that allows CmdExec.
2. Assign the developer's SQL login to use that proxy.
3. The developer can run only that step under that proxy.

Protection

They can only use that **single proxy identity**, not access the whole server.

◆ Scenario 5: Controlled Access for Support Teams

A support team should run SQL Agent jobs that:

- Restart services
- Move files
- Run PowerShell diagnostic scripts

But you don't want to give them:

- RDP access
- Local admin rights

Solution

Use Proxies to give them **just enough permissions** to run jobs.

🔶 Scenario 6: Data Export to CSV in Restricted Directories

A job exports data using BCP:

bcp "select * from Sales" queryout "\\fileshare\clientA\sales.csv" -c

SQL Agent cannot write to that share.

Solution

Use a Proxy with CmdExec rights pointing to an account with folder permissions.

🧩 How Proxies Work Internally

Steps:

1. Create a **Credential** (stores Windows account & password).
2. Create a **Proxy** that uses that Credential.
3. Assign Proxy to one or more **Subsystems** (e.g., CmdExec).
4. Allow selected **SQL Logins** to use the proxy.
5. In the SQL Agent job step → choose **Run As** = the Proxy.

Execution Flow:

Job Step → Proxy → Credential → Windows Account → Executes Command

🔑 Security Best Practices

- Do **NOT** use a domain admin account for proxies.
- Create **dedicated least-privilege domain accounts**.
- Use **strong passwords** and rotate them.
- Keep proxies restricted to necessary subsystems.
- Monitor job history for failed logins indicating password changes needed.

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🧪 Example: Creating a Proxy (T-SQL)

```
-- Step 1: Create Credential
CREATE CREDENTIAL BackupCopyCred
WITH IDENTITY = 'Domain\BackupUser',
SECRET = 'StrongPassword!123';

-- Step 2: Create Proxy
EXEC msdb.dbo.sp_add_proxy
   @proxy_name = 'BackupFileCopyProxy',
   @credential_name = 'BackupCopyCred';

-- Step 3: Allow CmdExec subsystem
EXEC msdb.dbo.sp_grant_proxy_to_subsystem
   @proxy_name = 'BackupFileCopyProxy',
   @subsystem_id = 3;  -- CmdExec

-- Step 4: Allow SQL Login to use proxy
EXEC msdb.dbo.sp_grant_login_to_proxy
   @proxy_name = 'BackupFileCopyProxy',
   @login_name = 'AppUser';
```

Summary

Feature	Explanation
Purpose	Run job steps with different permissions than SQLAgent service
Uses	SSIS, PowerShell, CmdExec, File system, Network shares
Security	Least privilege, controlled access
Who Uses It	DBA, ETL developers, infrastructure teams, automated tasks

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